Introduction to Information Security: Course Plan

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- *"Welcome! Before we begin, let's sort a couple of things."*

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Particulars

- Credits: 2
- Coverage: State-of-the-art of information security
- Language: Content delivery, exams, etc. will be in English
- Schedule: Spring semester (~~Monday & Thursday 15:35-17:00~~)
  - https://outlook.office365.com/owa/calendar/5e0c4525d65a48aaabcb57ec4e52e72c@iiit.ac.in/c5b36969693449968 1af10fc7dc4e04d1732910326504086 6816/calendar.html
  - https://outlook.office365.com/owa/calendar/5e0c4525d65a48aaabcb57ec4e52e72c@iiit.ac.in/c5b36969693449968 1af10fc7dc4e04d1732910326504086 6816/calendar.ics

Course contents

- Concepts of information security across different platforms
  - Cryptography
  - OS and network security
  - Web security
  - Mobile security
  - ML model security

PS: Course contents may vary depending on the interest and response from the students

Course contents

- Concepts of information security across different platforms
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- Student proficiency survey/form
  - *https://tinyurl.com/iisproficiencyform*
    - Deadline: **January 07, 2026, 11:59 PM**

PS: Course contents may vary depending on the interest and response from the students

Grading, examinations, and policies

- Grading
  - Relative grading
  - Final grade set: {A, A-, B, B-, C, C-, D, F} | No ‘P’ grade possible

- Examinations & marks distributions
  - Quiz, assignment : 40% (20 Marks)
  - Mid-term exam : 60% (30 Marks: 20+10 split)

- Policies
  - Do and submit your own original work
  - Cheating, plagiarism, and academic dishonesty will carry consequences

Textbooks and references

- There is no single textbook that can cover entire course material, some good references to start are given below (most of them are available on the Internet)

1. W. Stallings. “Cryptography and Network Security: Principles and Practice.”
2. C. Kaufman, R. Perlman, and M. Speciner. “Network Security: Private Communication in a Public World.”
3. Research papers.

Setting expectations

Some **REALLY** important lessons!

- *"Finally, prioritize things!"*

Course contents

- Concepts of information security across different platforms
  - Cryptography
  - OS and network security
  - Web security
  - Mobile security
  - **ML model security**

+ Security Basics
- Overview of security terminologies
- Basics of systems/network security
- Types of attacks
- Basics of cryptography
- Overview of software vulnerabilities
- Practices in cybersecurity

+ Introduction to Machine Learning
- Introduction to statistics and machine learning
- Basic workflow and terminologies in ML
- Mathematical formulation of problems in ML
- Loss functions and gradients
- Optimisation (backpropagation), regularisation, and metrics
- Image-related tasks and CNNs

+ Introduction to Security in Machine Learning Systems
- Taxonomy of ML security
- Threat models in ML security
- Types of watermarking, Uchida's algorithm

+ Attacks on DNNs
- Backdoor attacks
- Model extraction attacks
- Adversarial examples

+ Defences and Ownership
- Backdoor detection and removal
- Defences against model extraction
- Non-trivial Ownership
- Setting Thresholds
- eXplainable AI (XAI)
- False Claims

PS: Course contents may vary depending on the interest and response from the students

The Real Stars of the Course

Aaryan Ajay Sharma
BTech+MS Student

Aditya Kulkarni
BTech+MS Student

Raghav Doshi
BTech+MS Student

Aryanil Panja
BTech+MS Student

Venkata Sai Pranav Bachina
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Good luck.

Thank you!